

Practical Lab 2 Answers

January 20, 2026

- Go to the UCSC genome browser (genome.ucsc.edu).
- Use assembly Human Dec. 2013 (GRCh38/hg38)
- Enter ALDH2 into the search field and hit Return
- Select one of the RefSeq entry link [ALDH2 at chr12:11766933-111817531](#)
- Show the tracks listed below, hide all others and Refresh

Track	Name	Setting
Mapping and Sequencing	Base Position	Dense
Genes and Gene Prediction	NCBI RefSeq	Full
Phenotype and Literature	GWAS Catalog	Dense
Expression	GTEx Gene	Full
Regulation	ENCODE Regulation	Show
Variation	CommonSNPs (150)	Dense

Go to Genes and Gene Predictions, click on NCBI RefSeq and configure the NCBI RefSeq track to show only **RefSeq Curated** as shown and hit submit.

Webb - HGEN-502-002-2018S: X Reincarnation Blues by Micha X NCBI RefSeq Track Settings X

genome.ucsc.edu/cgi-bin/hgTrackUi?hgsid=710220

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NCBI RefSeq Track Settings

RefSeq gene predictions from NCBI - Annotation Release NCBI Homo sapiens Annotation Release 109 (2018-03-29) ([All Genes and Gene Predictions tracks](#))

Maximum display mode: full Submit Cancel Reset to defaults

Label: ☒ gene symbol ☐ accession ☐ OMIM ID

Select views ([help](#)):

NCBI RefSeq full UCSC RefSeq dense

List subtracks: ☐ only selected/visible ☒ all (1 of 7 selected)

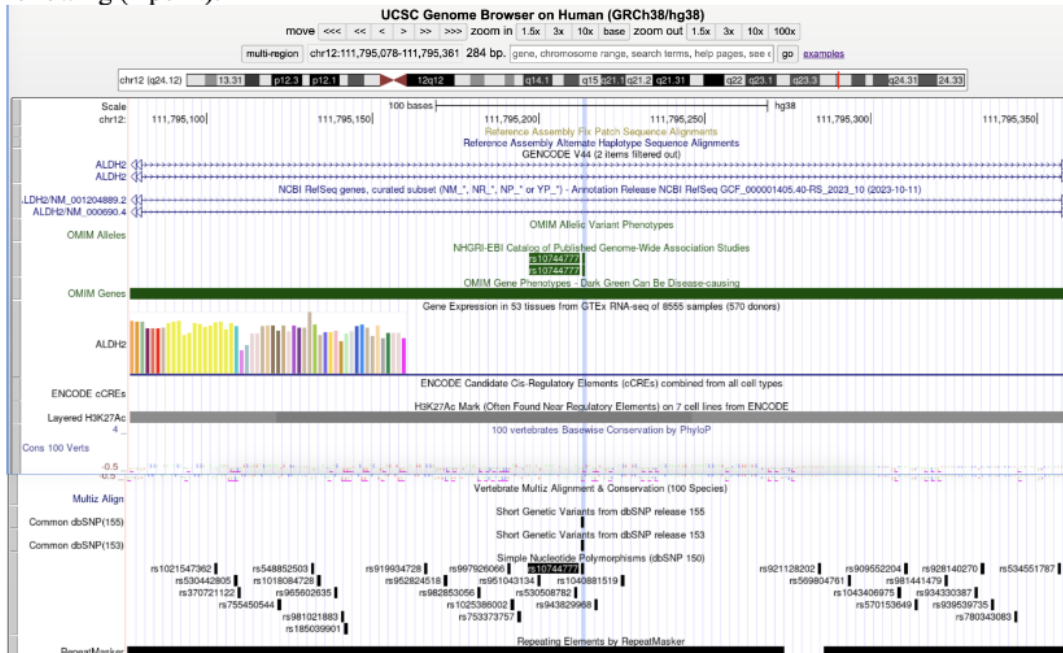
☐ full	RefSeq All	NCBI RefSeq genes, curated and predicted sets (NM_*, XM_*, NR_*, XR_*, and YP_*)	schema
☒ full	RefSeq Curated	NCBI RefSeq genes, curated subset (NM_*, NR_*, and YP_*)	schema
☐ full	RefSeq Predicted	NCBI RefSeq genes, predicted subset (XM_* and XR_*)	schema
☐ full	RefSeq Other	NCBI RefSeq other annotations (not NM_*, NR_*, XM_*, XR_*, or YP_*)	schema
☐ full	RefSeq Alignments	RefSeq Alignments of RNAs	schema
☐ full	RefSeq Diff	Differences between NCBI RefSeq Transcripts and the Reference Genome	schema
☐ dense	UCSC RefSeq	UCSC annotations of RefSeq RNAs (NM_* and NR_*)	schema

1 of 7 selected

Submit

Data version: NCBI Homo sapiens Annotation Release 109 (2018-03-29)

Take a screenshot of the genome browser after configuring tracks. Attach with answers to the following (1 point):



1. How many RefSeq transcripts are shown for ALDH2 (1 point)?

There are 2 transcripts for this gene.

2. Why is more than one transcript shown (1 points)?

Because of alternative splicing

2. How much genomic DNA in basepairs does the ALDH2 RefSeq transcript with the most exons cover (2 points)?

50600

Position: hg38 chr12:111,766,933-111,817,532 Size: 50,600 Total Exon Count: 13 Strand: +

3. How many basepairs is the ALDH2 RefSeq transcript with the most exons (Use GenBank ID NM_000690.4) (2 points)?

9561 bp

5. Is the direction of transcription of *ALDH2* from centromere to telomere or from telomere to centromere (2 points)?

Centromere to telomer

6. In the Association Studies track SNP rs10744777 has been associated with what phenotype (2 points)?

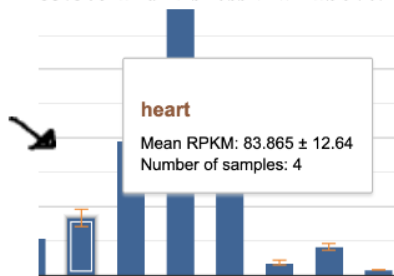
Ischmeric Stroke (Small artery occlusion)

7. What is the tissue that *ALDH2* is highest expressed from the GTEx RNAseq project and the level of expression (2 points)?

594.94 RPKM in Liver

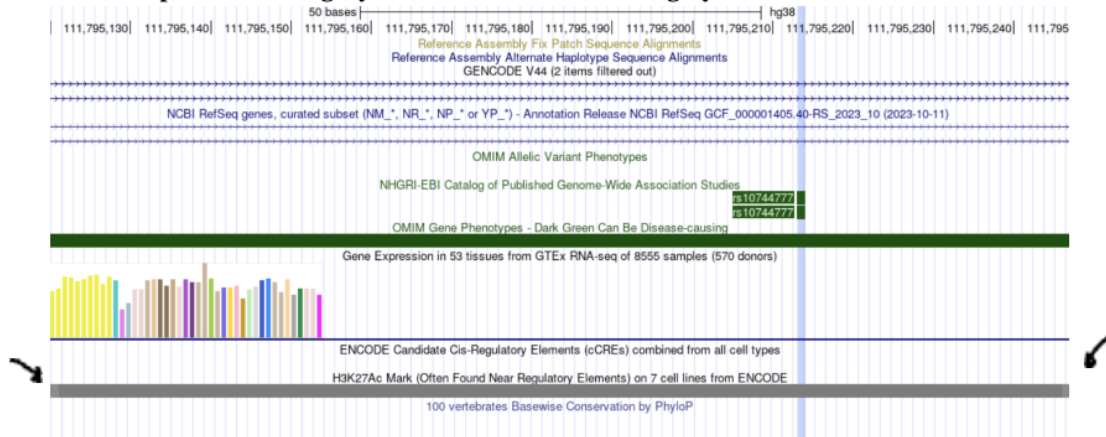
8. What is the level of expression of *ALDH2* in the heart from the HPA RNA-seq normal tissue found at NCBI (<https://www.ncbi.nlm.nih.gov/>)? Is it greater or less than the expression above (2 points)?

83.865 and it's less than above.



9. Go back to UCSC and look at the “H3K27Ac Mark (often found near active regulatory elements) on 7 cell lines from ENCODE. In what region of the *ALDH2* gene is there the best evidence for an active regulatory elements (2 points)?

(chr12 :111.795.122-111.795.249) I found this on the map , Between these two that showed on the map with darker gray. It is between intron 1 roughly.



10. Are there any common nonsynonymous SNPs in *ALDH2* (1 point)?

No

11. In what pathway is the *ALDH2* gene involved (2 points)?

On KEGG-Kyoto Encyclopedia of Genes and Genomes : the pathway that *ALDH2* gene is involved : Glycolysis/Gluconeogenesis

Aldehyde dehydrogenase,oxidative pathway and alcohol metabolism